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Enhancing Sign Language Learning Through Augmented Reality

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Abstract

Sign language education often relies on in-person instruction, limiting accessibility for learners without access to trained professionals. This project investigates a vision-based system for British Sign Language (BSL) learning that delivers real-time feedback in an augmented reality (AR) environment. The proposed solution integrates hand landmark detection with neural network-based static gesture classification to support interactive self-guided practice. Current development focuses on recognising a subset of single-handed BSL alphabet signs, with upcoming work extending toward dynamic gestures using temporal models. Planned evaluation will measure recognition accuracy and user performance improvements through iterative testing with representative learners. The anticipated contribution of this work is a framework that demonstrates the feasibility of AR-assisted sign language learning and identifies key technical and pedagogical challenges in deploying accessible signing support tools at scale.

1 Introduction

Sign languages are essential for the Deaf and hard-of-hearing community, supporting accessible communication, social inclusion, and self-advocacy. However, access to formal British Sign Language (BSL) learning remains limited due to shortages of qualified instructors and the cost and scheduling constraints associated with in-person training. As a result, many learners rely on self-study resources such as diagrams or video tutorials, which often fail to provide real-time correction or support accurate movement learning [12]. Without immediate feedback on handshape, spatial placement, or temporal motion, beginners frequently develop errors that hinder progress and confidence.

Advances in computer vision (CV), machine learning (ML), and augmented reality (AR) provide new opportunities to overcome these challenges. AR encourages embodied learning by overlaying guidance directly within the user's physical environment, and recent work demonstrates improvements in retention and motivation when learners receive interactive and personalised support [12, 24]. Simultaneously, large-scale datasets such as BSL-1K [2] and multimodal corpora like How2Sign [11] have driven progress in automatic sign recognition using spatial-temporal modelling.

Despite these advances, deployable learning systems remain rare. Many high-performance recognition models rely on GPU acceleration or controlled recording environments, making them unsuitable for widely accessible educational use [9]. Furthermore, existing tools often support only static pose matching and lack temporal understanding, which is essential because subtle motion differences can change meaning entirely in sign languages [13]. This gap highlights the need for systems that combine strong recognition performance with intuitive instructional feedback in real-time.

This project proposes a lightweight, browser-based AR learning assistant capable of operating cross-platform without installation barriers. The system uses MediaPipe Hands for robust landmark tracking, a compact neural classifier for static one-handed BSL alphabet recognition, and interactive visual feedback delivered within an AR interface. The long-term objective extends toward dynamic movement assessment using temporal alignment approaches such as Dynamic Time Warping (DTW). Through this research, we aim to design and evaluate a scalable and inclusive technological solution that improves engagement, accuracy, and accessibility in BSL learning.

2 Problem Statement

Current BSL learning tools do not provide the real-time, error-corrective feedback necessary for effective independent acquisition of sign accuracy and fluency. Traditional resources lack personalisation, and existing automated systems often struggle with varied environments, generalising across different users, and recognising temporally-defined signs [9]. Moreover, many CV-based recognition solutions are not optimised for consumer-grade hardware or web deployment, restricting accessibility outside specialist settings.

There is a clear need for an interactive learning system that is able to:

- operate fully in-browser using efficient inference,
- guide beginners by detecting and explaining errors in static hand configurations, and
- eventually support dynamic gesture interpretation through temporal modeling.

This project addresses these needs by developing and evaluating a real-time AR-assisted BSL tutoring assistant that integrates lightweight ML-based hand-pose classification with interpretable instructional feedback. The intended contribution is a deployable prototype demonstrating that modern hand-tracking technologies and compact deep learning models can effectively support accessible sign language instruction for non-expert learners.

3 Background Survey

Sign languages are complex visual gestural communication systems with unique expressive styles, educational rules, and grammatical structures [22]. British Sign Language (BSL) relies on coordinated gestures, spatial positioning, orientation, movement, facial expressions, and body posture to convey linguistic meaning. These multichannel characteristics pose challenges for automated recognition due to occlusions, articulation uncertainty, and strong signer-dependent variation [1]. Even though the field of Sign Language Recognition (SLR) has advanced significantly, the majority of systems still find it difficult to function reliably in real-world learning environments where users lack controlled backgrounds or consistent gesture form.

3.1 Vision Based Sign Language Recognition

From early manual pipelines to modern deep learning algorithms, Vision based Sign Language Recognition (SLR) has advanced a long way. Traditional approaches often relied on skin colour segmentation, optical flow fields, and Histogram of Oriented Gradients (HOG) to extract motion cues from RGB video, followed by traditional machine learning algorithms such as Support Vector Machines (SVMs) or Hidden Markov Models (HMMs) [26]. Although these systems performed reasonably well in controlled settings, they performed poorly in real world variations, especially when exposed to barriers, crowded backgrounds, and various kinds of skin tones [3]. Furthermore, handcrafted pipelines did not prove durable to abnormal lighting circumstances and signer-specific syntax differences, which are inevitable in home learning contexts.

The shift toward deep learning has made it possible to extract hierarchical visual representations that are specific to gesture patterns. While 3D CNNs and two stream architectures directly incorporate spatial temporal motion dynamics from raw video frames, Convolutional Neural Networks (CNNs) have proven to perform well in static handshape identification [19, 23]. By incorporating long range temporal attention, transformer-based architectures have recently demonstrated state-of-the-art scalability in continuous SLR and sign-to-text translation tasks [8]. Large annotated datasets like BSL-1K [2], WLASL [17], and Phoenix2014T [13], that document variation in signing individuals, mobility patterns, and linguistic context, have emerged to support these deep models.

However, performance gains from end-to-end CNN/Transformer architectures come with substantial computational cost. Inference often depends on powerful GPUs or specialised accelerators due to high resolution inputs and large model sizes [23]. These constraints hinder deployability on mobile devices or in web based environments where learners expect real time responsiveness. Moreover, pixel based models remain sensitive to environmental noise such as camera angle shifts, sleeve color similarity, and partial occlusions, factors that are out of human control and can disrupt reliable sign recognition outside controlled settings [3].

The implementation of Lightweight skeletal models based on extracted 2D or 3D hand landmarks significantly reduces the computational load while improving generalisation to variabilities in skin tone and background conditions [18]. This encourages a shift away from raw video towards structured geometry suitable for efficient browser based machine learning, aligning closely with the goals of the current project.

3.2 Skeletal Modelling and Pose Feature Learning

Skeletal modelling has emerged as a dominant strategy in real-time sign recognition due to its efficiency and robustness compared to pixel based video analysis. Modern middleware such as MediaPipe Hands estimates 21 anatomically meaningful 2D/3D landmarks per hand at interactive frame rates [18], enabling structured representations of hand configuration independent of background texture, skin tone, or camera lighting. By operating on sparse geometry rather than dense images. Models avoid high-dimensional convolutional pipelines, resulting in reduced computational load and significantly improved flexibility to limited resource environments, such as smartphones and browsers.

Pose based descriptors further enable extraction of linguistically grounded features, such as joint rotation angles, inter-finger distances, palm orientation, and hierarchical kinematic constraints [25]. These characteristics are particularly valuable in sign languages where language correctness is defined by handshape, location, and movement relative to the body [6]. Angle based features capture internal movement (e.g., finger bending) that raw landmark coordinates fail to distinguish, improving recognition of minimal pairs and duplicate forms. Research suggests such features align more closely with perceived clarity for learners and sign-language instructors [22].

Enhanced generalisation across users and perspectives is a further advantage of skeletal modelling. Normalisation techniques including scale alignment, reference point anchoring (e.g., wrist as anatomical root), and mirroring compensation for left/right hand dominance can reduce noise caused by partial interruptions or lateral camera angles. These steps enable interpretable, domain specific inference without the need for extensive data augmentation.

From an engineering perspective, compact posture attributes provide high compatibility with web-based inference engines like WebAssembly and ONNX Runtime Web, which need predictable execution routes and minimal memory consumption for real-time responsiveness [21]. Furthermore, unlike deep CNN activations, joint level deviations can be visualised and communicated as instructional guidance, which is a crucial requirement for personalised tutoring systems [4]. The skeleton structure also directly assists with the explainability and feedback creation.

This project's emphasis on accessible, browser-based BSL learning is greatly supported by skeletal modeling's combination of linguistic relevance, computational efficiency, and interpretability qualities, which make it an ideal basis for sign language tutoring.

3.3 Temporal Dynamics and Motion Interpretation

Temporal modelling remains essential in sign language recognition because many signs are distinguished not only by their static handshapes but by the motion trajectory, speed, repetition, and start–end spatial locations [13, 27]. Signs that appear visually similar in a single frame can have entirely different meanings once temporal evolution is considered, such as the difference between polite BSL gestures like *"thank you"* and *"please"*, where motion direction and sppech length act as key differences [6]. Failure to integrate movement cues can therefore, lead to uncertainty and lower the overall learner feedback quality.

Modern computer vision models address temporal variability using recurrent based neural architectures. Long Short Term Memory (LSTM) networks capture sequential dependencies and have demonstrated strong performance for continuous SLR tasks [8]. More recently, Transformer architectures enable richer temporal context and global modelling of gesture sequences, achieving state-of-the-art recognition in large video corporations [8]. However, these approaches are computationally intensive and require large datasets to generalise effectively across different learners and signing styles.

Traditional positional techniques, such as Dynamic Time Warping (DTW), offer an alternative that is lightweight, data efficient, and particularly well suited to user driven educational feedback [20]. DTW allows temporal stretching and compression to account for natural signing variability while still preserving the linguistic structure of gestures. Because beginners often perform signs, either more slowly or with inconsistent timing compared to more fluent signers, DTW provides meaningful similarity scoring without imposing strict timing constraints. As such, DTW remains highly relevant for deployment-constrained environments and for real time tutoring systems, which must provide interpretable feedback. This project utilises DTW as a useful starting point for modelling motion based BSL gestures, allowing an AR learning interface to automatically assess temporal consistency.

3.4 Datasets for Sign Recognition and Learning

The advancement of automatic recognition has been greatly aided by the availability of large sign language datasets. The small size and difficult management of early datasets limited their usefulness to real-world generalisation. Scalable, diverse collections that capture natural signature variation have become the focus of recent academic projects. For instance, research into broad vocabulary classification was made possible by BSL-1K, which introduced more than 1,000 British Sign Language (BSL) sign classes derived from broadcast video [2]. For sign-to-speech translation and contextual gesture understanding, How2Sign offers a complete multimodal dataset that connects RGB video, depth streams, and motion capture [11].

Despite having advanced recognition systems, these datasets aren't specifically made for educational purposes. They usually include fluent signers rather than inexperienced learners, which limits the model's capacity to accept linguistic mistakes or uncertainty typical of early skill acquisition [**?**]. Additionally, many datasets are less suitable for feedback based correction or browser embedded landmark inference because they lack deep skeletal or linguistic annotations like handshape category and signer viewpoint. Alignment metadata, such as start and finish frames of important movement phases, is frequently lacking or inconsistent among datasets, even in cases when temporal dynamics are provided.

Another limitation relates to deployment and accessibility. Large multimodal datasets are difficult to implement in lightweight, WebAssembly-based educational systems since they usually require high bandwidth processing and storage [21]. Additionally, licensing limitations on broadcast video, like those seen in BSL-1K, can make it more difficult to integrate it into public facing applications, particularly those that facilitate large scale digital learning.

These gaps highlight the need for datasets specifically tailored for instructional goals rather than recognition accuracy alone. Specifically, future datasets should prioritise:

(1) inclusion of slow, learner appropriate signing styles,
(2) educationally grounded accurate labels,
(3) rich annotations for landmarks that are compatible with browser inference and,
(4) varied camera angles to reflect home learning conditions

Addressing these requirements would strengthen the bridge between academic SLR research and deployable real-time tutoring systems for new BSL learners.

3.5 Sign Language Learning Technologies

E-learning platforms for sign language often rely on pre-recorded videos or static diagrams, which provide no error correction or visual feedback [15]. However, learning research highlights the importance of embodied, feedback-driven rehe arsal for motor-skill development [16]. Augmented reality systems can therefore provide meaningful learning advantages by anchoring feedback in a user's own visual context [5].

Recent AR-assisted prototypes demonstrate improved motivation and gesture refinement through real-time correction [10, 24]. Yet, many rely on mobile GPUs or marker based tracking, diminishing accessibility. VR-based signing tutors [4] demonstrate immersive interaction but require specialised devices that limit adoption.

A persistent gap exists between high-performing recognition research and deployable, everyday sign language education tools. The system proposed in this research addresses this disconnect by:

- leveraging lightweight landmark-based feature extraction,
- integrating interpretable angle-based feedback aligned with sign phonology,
- utilising ONNX Runtime Web for browser-based, cross-device inference,
- and incorporating temporal scoring extensions for dynamic signing.

This positioning differentiates the proposed approach as both technologically feasible and educationally grounded, strongly aligned with the accessibility goals of inclusive digital learning.

3.6 Gaps and Opportunities

Sign Language Recognition (SLR) research has produced impressive results in controlled laboratory settings, however the conversion of these developments into real-world educational materials continues to be limited and scattered [3, 15]. Current systems frequently prioritise benchmark evaluation above learner experience, which creates a gap between the practical requirements of new learners pursuing self-guided study and the latest developments in recognition research.

There still exist multiple substantial gaps in the research and deployed technologies:

(1) **Lack of accessible learning platforms:** Deployment in browser-based or low-resource contexts is hindered by the majority of existing systems' reliance on native mobile applications or strong GPUs [21]. Many students who simply have access to standard consumer gadgets are forced excluded from this.
(2) **Limited meaningful real-time feedback:** Many prototypes, recognition output is limited to binary correctness judgements, with few details about *why* an error happened or *how* to improve [14]. According to educational studies, inexperienced students need educationally aligned input for them to develop proper hand shapes and motion patterns.
(3) **Insufficient support for dynamic gesture evaluation:** Due to their simpler modelling, isolated static signs continue to have a large bias; nevertheless, a significant number of meaningful indications depend on temporal motion, directional movement, or volume variation [13]. Systems that ignore temporal dynamics are unable to scale to cover the entirety of the BSL curriculum.
(4) **Poor adaptability under real-world changes:** When models are used outside of clean datasets, variations in signer anatomy, backdrop chaos, lighting, occlusion, and camera positioning reduce performance [2]. Household learning environments, where conditions cannot be strictly regulated, are rarely specifically addressed by systems.

Together, these shortcomings highlight a significant area for innovation: the creation of a fully browser-based BSL teaching system that can provide inexperienced students with thorough, responsive, and academically relevant feedback. The suggested approach attempts to close the gap between technical recognition

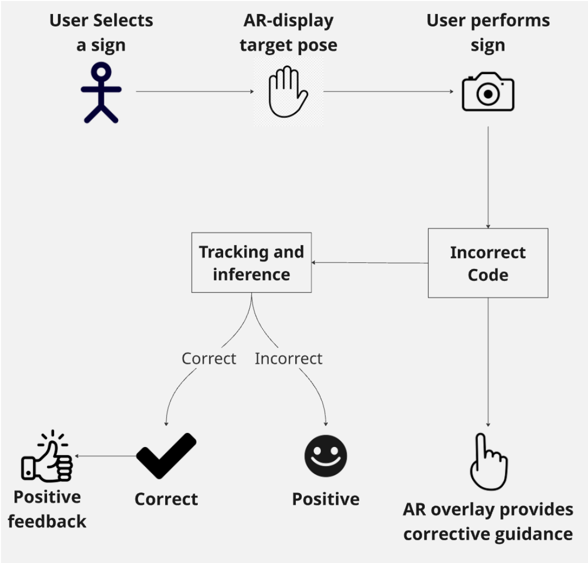

Figure 1: AR Interaction Flow Diagram

performance and successful language learning by utilising lightweight skeletal elements for fast inference and enhancing them with educational signals and temporal comparison techniques. This strategy promotes equal access to sign language learning and might significantly advance the Deaf community's digital inclusion.

4 Proposed Approach

The primary objective of this project is to provide an engaging, augmented reality-based British Sign Language (BSL) learning environment. Learners actively practise signs in front of their webcam instead of simply taking in educational content, and the technology delivers them immediate real-time feedback on hand pose accuracy, signing posture, and motion correctness. The AR interface encourages kinaesthetic learning through physical contact by superimposing visual guidance cues directly into the learner's field of view [5]. In a publicly accessible, browser-based setting, the system aims to mimic important elements of human coaching, including encouragement, organised growth, and constructive criticism. The end-to-end system architecture is illustrated in Figure 5, while the complete learner interaction lifecycle is summarised in the high-level AR interaction flow shown in Figure 1.

To realise this vision, the proposed implementation integrates lightweight machine learning models with real-time computer vision in an interactive learning application. The following illustrations highlight the essential elements within the proposed methodology.

4.1 Static Gesture Recognition via Hand Landmark Classification

For alphabetic and fixed pose signs, the system uses MediaPipe Hands to extract 21 hand landmarks in real time [18]. These 3D points are flattened into a 63 dimensional feature vector and classified using a compact fully connected neural network deployed through ONNX Runtime Web [21]. Lightweight inference ensures broad device compatibility without requiring GPUs or native software installation. Figure 2 summarises the data processing steps,

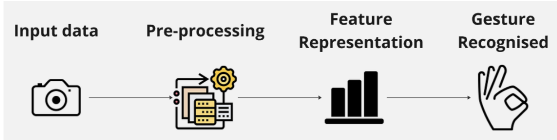

Figure 2: Data Processing & Feature Representation Diagram

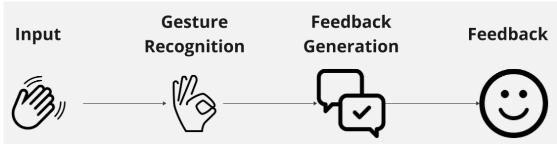

Figure 3: Feedback System Diagram

showing how raw MediaPipe landmarks are normalised, transformed, and fed into the neural classifier.

The system returns a confidence score indicating similarity between the user's posture and the target BSL sign, enabling clear and interpretable feedback. An example of the live practice interface and confidence scoring is shown in Figure 4

4.2 Temporal Gesture Assessment via Dynamic Time Warpping

Following similar techniques employed in lightweight recognition prototypes, the system utilises Dynamic Time Warping (DTW) to compare user motion sequences to generate patterns for dynamic signs incorporating motion trajectories (such as signalling gestures) [13, 29]. Sequences are evaluated using a normalised matching score after being resampled for temporal alignment.

Explanation:

- DTW enables a range of frequencies
- It performs well with small datasets
- Neural classifiers for motion encoded gestures are linked

4.3 Real Time Feedback Mechanism

A key component of learning sign language is receiving immediate corrective feedback. The proposed methodology assesses:

- Colour coded skeleton alignment indicators for pose correctness
- Text prompts for hand positioning (e.g. "raise palm", "move closer to camera")
- Confidence scores and predicted sign labels for immediate user awareness
- Motion progress indicators during dynamic sign attempts

This design supports self correction, more frequent practice, and stronger engagement of the user compared to traditional resources [14]. The flow of structured feedback, from detection to user facing guidance, is illustrated in Figure 3. The full practice workflow, from landmark input to model output, is shown in the training and deployment pipeline displayed in Figure 6.

4.4 Deployment and Accessibility Strategy

The application runs fully in the browser using WebAssembly acceleration, enabling:

- Zero install access across laptops, tablets, and mobile devices

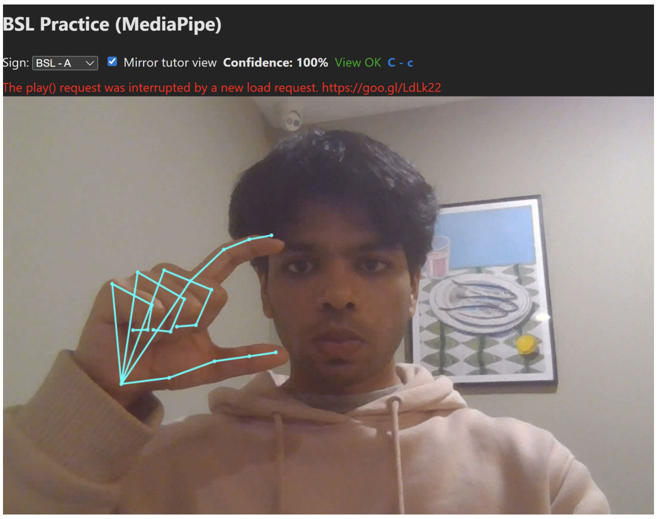

Figure 4: Early prototype of BSL practice interface showing MediaPipe hand detection and confidence scoring during sign execution.

- Privacy preservation by keeping all inference on device
- Instant scalability, no server compute or login required

This aligns with the project's commitment to inclusivity and universal access for diverse user groups.

4.5 Feasibility and Potential Risks

The significant integration of lightweight neural inference technologies and reliable 3D hand tracking pipelines makes the suggested technique physically possible. In real time gesture recognition systems, landmark based classification has proven to be a dependable method that maintains high accuracy while requiring little processing power. Because of this, browser based execution with ONNX Runtime Web is both practical and scalable on a variety of user devices.

It is also possible to score gestures dynamically using DTW, especially in applications where datasets can be limited. Learners can execute gestures with natural variation thanks to DTW's ability to deal with temporal variations like different signing speeds. Because of its integration, a wider range of BSL signs that rely on motion rather than just static handshape can be supported by the system.

But there are certain risks to be aware of. First, visual occlusions, low lighting, and hand dominance (left vs. right performance) can still affect system reliability and result in incorrect classification or low confidence scores. Second, the size of the current dataset restricts generalisation across various learners, signing styles, and demographic variations, a problem that is well known in the field of sign language recognition research [11]. Third, even though WebAssembly allows for high-performance on-device processing, low power devices may still face latency or lower frame rates, which could negatively impact user experience.

Progressive dataset growth (e.g., adding BSL-1K for better generalisation), verification procedures to accommodate left handed users, and dynamic performance adjustment based on real time browser capability detection can all help reduce these risks. The system's long-term scalability will be enhanced by ongoing iteration and community-driven data collection.

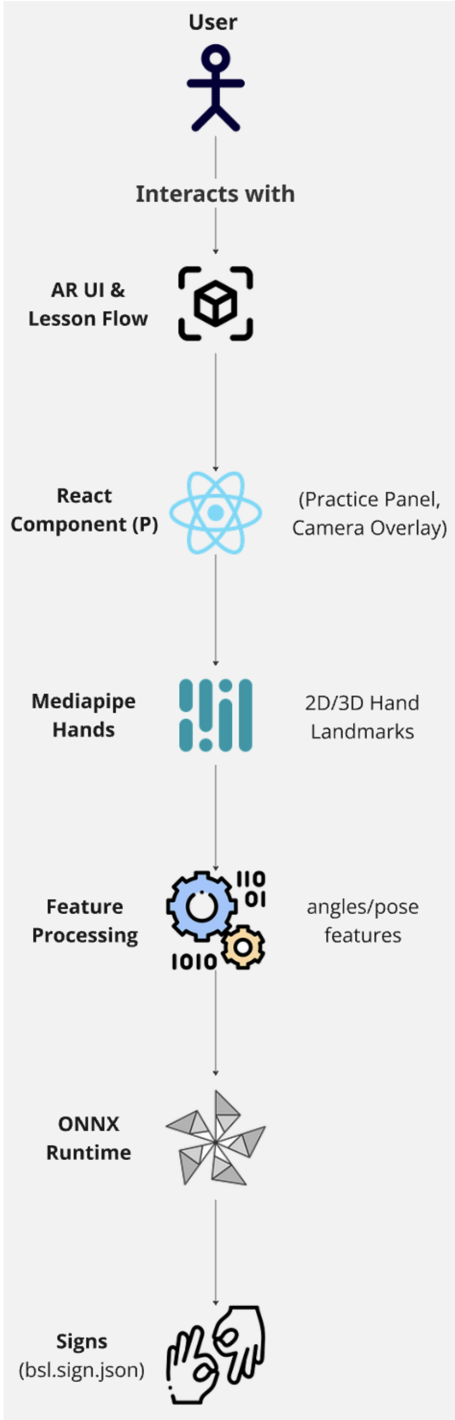

Figure 5: Overall System Architecture Diagram

5 Project Discussion

The current stage of development has demonstrated the potential of real time browser based gesture recognition as the foundation for an interactive British Sign Language learning tool. Initial functional prototypes confirm that lightweight neural models deployed via ONNX Runtime Web can classify static hand poses with low latency on consumer hardware, aligning with findings

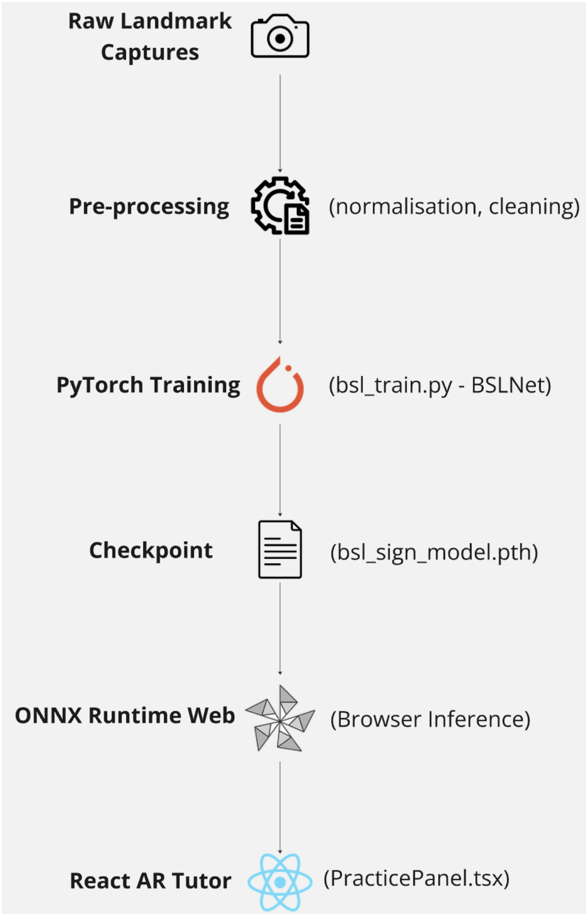

Figure 6: Training and deployment workflow for the BSL sign recognition model

from recent landmark-based learning systems [21, 25]. This validates the choice of on device inference and supports the project's accessibility objectives.

However, the early prototype also indicates significant challenges, mostly related to robustness and generalisation. Variations in user backgrounds, hand sizes, camera angles, and light conditions can have a substantial impact on detection accuracy because the current dataset is restricted to controlled capture sessions. This is consistent with the acknowledged limitations of sign-language technologies, where human variation and environmental noise continue to be major adoption hurdles [9, 11]. To facilitate wider use, it will consequently be crucial to deploy augmentation techniques and improve dataset diversity.

The use of MediaPipe Hands has provided highly efficient landmark estimation, although sensitivity to occlusion and camera location is still a limitation. In order to address these problems, the system uses heuristics through a process called "view gating," which verifies hand visibility and pose orientation prior to identification and encourages students to sign correctly. This strategy minimises false negative results that might discourage users while promoting scientific accuracy [24].

From a learning perspective, qualitative feedback has already shown that users have a good understanding of visual overlays and confidence ratings. Immediate feedback has demonstrated to being able to accelerate procedural skill development in embodied learning scenarios. Additionally, compared to static resources like video lessons [12], the AR-based interaction seems to offer a more engaging and gamified learning experience. These initial responses support the educational relevance of the proposed approach.

Dynamic gesture evaluation, which is currently in the prototype stage and uses Dynamic Time Warping (DTW), offers a promising method for motion-based sign assessment but adds more variability to user performance. DTW works well in low-data settings, but if the motion vocabulary increases sufficiently, switching to neural temporal models may be required [8, 20]. This creates a trade off between temporal categorisation accuracy and computing efficiency in the future.

In summary, the project is advancing forward according to plan and shows early educational and technological potential. Expanding dataset coverage, strengthening UI-driven assistance for signing consistency, refining dynamic sign processing, and carrying out structured user evaluation to verify learning impact are the upcoming development milestones. These outcomes will help ensure that the final system not only recognises signs accurately but also serves as an effective, inclusive tool for independent BSL learning.

6 Work Plan

The remaining work will focus on improving system performance, expanding gesture capabilities, and carrying out a qualitative evaluation of the learning process. Continuous development cycles are used in this work to ensure that educational value and user involvement drive technical advancements [28].

6.1 Milestones and Deliverables

The project will be delivered through the following key phases:

- **Phase 1 — Dynamic Gesture Module Enhancement (January–February 2025)**
  If limitations on performance continue, enhance temporal recognition via assessing alternative neural sequence models and boosting DTW scoring [8, 20]. Expand motion based BSL vocabulary according to its educational relevance.
- **Phase 2 — AR Interface and User Interaction Design (February-March 2025)**
  Add progress monitoring, gamified reinforcement, and structured visual support (such as trajectory overlays and hand orientation prompts) to encourage motivation and self-regulated learning [4, 12].
- **Phase 4 — Evaluation and Usability Study (March 2025)**
  Conduct user testing with an emphasis on effectiveness, accessibility and error recovery. Task accuracy, learning progress, and subjective usability ratings like the System Usability Scale (SUS) will all be measured. [7].
- **Phase 5 — Thesis Write Up and Final Artefact Refinement (March-April 2025)**
  Integrate evaluation findings into the paper, finalise deployment, and prepare presentation materials for submission.

6.2 Research Tools and Risk Mitigation Approach

Continuous improvement will be guided by agile principles, and each milestone will consist of a performance evaluation and an opportunity to redesign if outcomes fall short of expectations.

If possible, technical and assessment tasks will be completed concurrently to reduce schedule risk. Progressive testing and backup techniques, including reduced gesture sets in case of time restrictions, will be used to mitigate dataset and performance risks that were previously identified (Section 5).

6.3 Expected Outputs

The planned deliverables include:

(1) A fully operational AR based BSL learning web application
(2) A validated lightweight BSL recognition pipeline
(3) A reproducible dataset + trained model artefact
(4) A fully written paper documenting contributions, evaluation, and impact

This well structured plan ensures the project will be delivered within the allocated timeframe while preserving research quality and user centred learning outcomes.

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